

The conjugation degree on the transcendental locus from the ground up

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Abstract

A ground-up account of the conjugation degree $\delta(z) = [\overline{\mathbb{Q}}(z, \bar{z}) : \overline{\mathbb{Q}}(z)]$ on $\mathcal{T} = \mathbb{C} \setminus \overline{\mathbb{Q}}$, written to be formalized. Every statement below is marked with its status in the Lean development at <https://github.com/carlok/moebius-transcendental-lean>: either the name of the declaration that proves it, or an explicit note that it is stated and not proved. No claim is made for any statement not so marked.

1 Setting

Write $\overline{\mathbb{Q}} \subset \mathbb{C}$ for the field of algebraic numbers and

$$\mathcal{T} := \mathbb{C} \setminus \overline{\mathbb{Q}}$$

for the transcendental locus. Let $\widehat{\overline{\mathbb{Q}}} = \overline{\mathbb{Q}} \cup \{\infty\}$.

Status. $\overline{\mathbb{Q}}$ is `Qbar`, defined as `(algebraicClosure Q C).toSubfield`, with `mem_Qbar_iff` identifying membership with `IsAlgebraic Q`. $\mathcal{T}$ is `Transcendentals`. Both proved.

Let $\Gamma = \mathrm{PGL}(2, \overline{\mathbb{Q}})$ act on $\widehat{\overline{\mathbb{Q}}}$ by $z \mapsto (az + b)/(cz + d)$ with $a, b, c, d \in \overline{\mathbb{Q}}$, $ad - bc \neq 0$, and let $J(z) = \bar{z}$. Put $\Gamma^\pm = \langle \Gamma, J \rangle$; its elements outside Γ are $z \mapsto (a\bar{z} + b)/(c\bar{z} + d)$ with algebraic coefficients.

2 The invariant

Definition 2.1. For $z \in \mathbb{C}$, the *conjugation degree* is

$$\delta(z) := [\overline{\mathbb{Q}}(z, \bar{z}) : \overline{\mathbb{Q}}(z)] \in \mathbb{N} \cup \{\infty\},$$

the degree of $\bar{z}$ over the field generated by $\overline{\mathbb{Q}}$ and z .

Status. `conjDegree`, valued in `ENat`, that is $\mathbb{N} \cup \{\infty\}$, defined as the degree of the minimal polynomial of $\bar{z}$ over $\overline{\mathbb{Q}}(z)$ when that element is integral, and $\top$ otherwise. The field $\overline{\mathbb{Q}}(z)$ is `gen`. The strata $\mathcal{T}_n = \{z \in \mathcal{T} : \delta(z) = n\}$ are `stratum`. Definitions only; the results of §3 are stated but not yet proved.

Remark. The definition is by minimal polynomial rather than by `Module.finrank`. Two reasons: it avoids the intermediate-field tower instances, and $\top$ is a genuine value where `finrank` would return the junk value 0 for an infinite degree, silently merging $\mathcal{T}_\infty$ with the empty case.

3 Elementary consequences

Everything in this section is elementary; it is recorded because the formalization needs it, not because it is difficult.

Lemma 3.1 (No holomorphic stabilizers). *If $g \in \Gamma$ is not the identity, every fixed point of g lies in $\widehat{\mathbb{Q}}$.*

Proof. Write $g(z) = (az + b)/(cz + d)$ with $a, b, c, d \in \overline{\mathbb{Q}}$. A finite fixed point satisfies $cz^2 + (d - a)z - b = 0$, a polynomial with coefficients in $\overline{\mathbb{Q}}$, not identically zero since $g \neq \text{id}$; so $z \in \overline{\mathbb{Q}}$. The remaining projective point is $\infty \in \widehat{\mathbb{Q}}$. $\square$

Proposition 3.2 (Transcendence is preserved). *If $z \in \mathcal{T}$ and $g \in \Gamma^\pm$ then $g(z) \in \mathcal{T}$.*

Proof. Suppose $g \in \Gamma$ and $g(z) \in \overline{\mathbb{Q}}$. Then $g^{-1} \in \Gamma$ has algebraic coefficients and $z = g^{-1}(g(z)) \in \overline{\mathbb{Q}}$, contradicting $z \in \mathcal{T}$. For $g \notin \Gamma$ write $g = h \circ J$ with $h \in \Gamma$; complex conjugation preserves $\overline{\mathbb{Q}}$, hence preserves $\mathcal{T}$, and the holomorphic case applies to h . $\square$

Proposition 3.3 (δ is $\Gamma^\pm$ -invariant). *$\delta(g(z)) = \delta(z)$ for every $z \in \mathcal{T}$ and $g \in \Gamma^\pm$.*

Proof. Let $g \in \Gamma$ with coefficients $a, b, c, d \in \overline{\mathbb{Q}}$. Since g and g^{-1} have algebraic coefficients, $\overline{\mathbb{Q}}(g(z)) = \overline{\mathbb{Q}}(z)$. Conjugating the defining expression gives $\overline{g(z)} = \bar{g}(\bar{z})$, where $\bar{g}$ has the conjugated coefficients $\bar{a}, \bar{b}, \bar{c}, \bar{d}$, again in $\overline{\mathbb{Q}}$ because $\overline{\mathbb{Q}}$ is conjugation-stable. Hence $\overline{\mathbb{Q}}(g(z), \overline{g(z)}) = \overline{\mathbb{Q}}(z, \bar{z})$, and the two degrees agree. For $g = J$ the two generators are exchanged and the field is unchanged. $\square$

Proposition 3.4 (Degree one). *For $z \in \mathcal{T}$, $\delta(z) = 1$ if and only if $\bar{z} \in \overline{\mathbb{Q}}(z)$.*

Proof. Immediate from Definition 2.1: a degree-one extension is trivial. $\square$

Remark (Why degree one is the interesting stratum). If $z\bar{z} = \rho$ with $\rho \in \overline{\mathbb{Q}}$ and $z \neq 0$, then $\bar{z} = \rho/z$ lies in $\overline{\mathbb{Q}}(z)$, so $\delta(z) = 1$. Complex conjugation on $\overline{\mathbb{Q}}(z)$ is then determined by the field operations rather than being independent data. That case is formalized separately, machine-checked and kernel-verified, at <https://github.com/carlok/diaz-modulus-lean>.

4 Multiplicity on a plane curve

For the target of §5 one needs the multiplicity of a point on a plane curve. Mathlib has no such notion; it is assembled here from `MvPolynomial.homogeneousComponent`.

Definition 4.1. For $P \in K[X, Y]$ and $p \in K^2$, let P_p denote P translated so that p is the origin. The *multiplicity* of p on $P = 0$ is

$$\text{mult}_p(P) := \min\{n : \text{the degree-}n \text{ homogeneous component of } P_p \text{ is non-zero}\}.$$

Status. `shiftTo` and `mult`. Proved.

Lemma 4.2. *If $P \neq 0$ then the set in Definition 4.1 is non-empty, so the minimum exists; and $P_p \neq 0$.*

Proof. A polynomial is the sum of its homogeneous components, so if all of them vanish then so does the polynomial; that gives the first claim. Translation by p is inverted by translation by $-p$, so it is injective, giving the second. $\square$

Status. `nonempty_of_ne_zero` and `shiftTo_ne_zero`. Proved, sorry-free.

5 The target, stated and not proved

Theorem 5.1 (Conjugation degree is circular degree). *Let V be an absolutely irreducible plane curve over $\overline{\mathbb{Q}} \cap \mathbb{R}$ whose real locus contains a transcendental point, let $\overline{V} \subset \mathbb{P}_{\overline{\mathbb{Q}}}^2$ be its projective closure of degree d , and let $m = \text{mult}_{I_+}(\overline{V}) = \text{mult}_{I_-}(\overline{V})$ at the circular points $I_{\pm} = [1 : \pm i : 0]$, zero when the point is absent. Then every $z = x + iy \in V(\mathbb{R}) \cap \mathcal{T}$ satisfies*

$$\delta(z) = d - m.$$

Status. Not proved, and not currently provable in Lean. The argument requires Bézout’s theorem for plane curves, the normalization of a projective curve, and the identification of the degree of a morphism with a function-field extension degree. As of Mathlib 81a5d257, none of the three is available: the `Bezout` declarations in Mathlib are `IsBezout`, the ring-theoretic class asserting that finitely generated ideals are principal, and there is no `intersectionMultiplicity`. The statement is recorded here so the formalization has a target and so the gap is visible.

Remark (A shortcut that does not work). It is tempting to read m off the leading form of P , as the multiplicity of $x^2 + y^2$ as a factor of the degree- d part. That quantity is the intersection multiplicity of $\overline{V}$ with the line at infinity, which bounds $\text{mult}_{I_+}(\overline{V})$ from above and equals it only when the branch is transverse. It agrees in the cases one checks by hand — a circle has leading form $x^2 + y^2 = (y - ix)(y + ix)$, so $d = 2$, $m = 1$; a real line has $d = 1$, $m = 0$ — because both are smooth at infinity. It is not a proof.

6 Status

Statement	Lean declaration	Module	Status
$\overline{\mathbb{Q}}$	<code>Qbar, mem_Qbar_iff</code>	<code>Basic</code>	proved
$\mathcal{T}$	<code>Transcendentals</code>	<code>Basic</code>	proved
$\overline{\mathbb{Q}}(z)$	<code>gen</code>	<code>Basic</code>	proved
Def. 2.1	<code>conjDegree</code>	<code>Basic</code>	defined
$\mathcal{T}_n$	<code>stratum</code>	<code>Basic</code>	defined
Lem. 3.1	—	—	stated only
Prop. 3.2	—	—	stated only
Prop. 3.3	—	—	stated only
Prop. 3.4	—	—	stated only
Def. 4.1	<code>shiftTo, mult</code>	<code>Multiplicity</code>	proved
Lem. 4.2	<code>nonempty_of_ne_zero, shiftTo_ne_zero</code>	<code>Multiplicity</code>	proved
Thm. 5.1	—	—	not provable yet

Five declarations are proved and `sorry`-free. Two are definitions with no results yet attached. Four elementary statements are written down but not formalized. One theorem is out of reach and the reason is named.

Use of AI assistance

The mathematics, this note and the Lean development were produced with substantial assistance from large language models, directed and accepted by the author, who is responsible for the content. Every declaration marked proved above is machine-checked by Lean 4 at the pinned toolchain `leanprover/lean4:v4.32.0`; no statement marked proved rests on an assertion by a model, and no statement not so marked is claimed.